MATH203 – APPLIED LINEAR ALGEBRA
HOMEWORK 4

Your Name: ____

Your Student Number: __________________________________

Your Tutorial TA: _____________________________________

Instructions.

(1) Show your work and justify your steps on every question.
(2) If a True/False statement is false, give a counterexample.
(3) You may use course materials, but your submitted solution must be written independently.
(4) Whenever a final answer is a concrete matrix, write the matrix entry-wise. A final answer left only as a product of matrices receives no credit, unless the question explicitly asks for a factorization such as a diagonalization, SVD, or MM^T form.

1. (20 points)

Answer True or False. If False, give a counterexample.

1.1) If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable, meaning it can be written as $A = PDP^{-1}$ for some diagonal matrix D.

1.2) If A is positive definite, meaning $x^T Ax > 0$ for every nonzero real vector x, then every diagonal entry a_{ii} is positive.

1.3) If A is Hermitian, meaning $A = A^H$, then the singular values of A, meaning the square roots of the eigenvalues of $A^H A$, are the absolute values of the eigenvalues of A.

1.4) If A is diagonalizable over $\mathbb{R}$, meaning $A = PDP^{-1}$ with real matrices P, D, and all eigenvalues are real, then A is symmetric, meaning $A = A^T$.

1.5) If A and B are both positive definite, meaning $x^T Ax > 0$ and $x^T Bx > 0$ for every nonzero real vector x, then $A + B$ is positive definite.

1.6) If A is real and skew-symmetric, meaning all entries of A are real and $A^T = -A$, then $I + A$ is invertible, meaning $(I + A)^{-1}$ exists.

1.7) If X is skew-Hermitian, meaning $X^H = -X$, then

$$U = (I + X)(I - X)^{-1}$$

is unitary, meaning $U^H U = I$.

1.8) If A is normal, meaning $AA^H = A^H A$, then eigenvectors corresponding to distinct eigenvalues are orthogonal, meaning their Hermitian inner product is zero.

1.9) If A is diagonalizable, meaning $A = PDP^{-1}$ for some diagonal matrix D, and all eigenvalues of A are positive real numbers, then A is positive definite, meaning $x^T Ax > 0$ for every nonzero real vector x.

1.10) If A is real, meaning all entries of A are real, and has a non-real eigenvalue $a + bi$ with $b \neq 0$, then $a - bi$ is also an eigenvalue of A.

Solution: **Problem 1.**

item	answer
1.1	True
1.2	True
1.3	True
1.4	False
1.5	True
1.6	True
1.7	True
1.8	True
1.9	False
1.10	True

$$\text{1.4 counterexample:} \quad A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}.$$

This matrix is diagonalizable with real eigenvalues $1, 2$, but it is not symmetric.

$$\text{1.9 counterexample:} \quad A = \begin{pmatrix} 1 & 100 \\ 0 & 2 \end{pmatrix}.$$

This matrix is diagonalizable with positive real eigenvalues $1, 2$, but

$$\begin{pmatrix} 1 & -1 \end{pmatrix} A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 - 100 + 2 < 0.$$

1.6 proof:

If $A^T = -A$ and $(I + A)x = 0$, then

$$0 = x^T (I + A)x = x^T x + x^T Ax = x^T x,$$

because $x^T Ax = 0$ for skew-symmetric A. Hence $x = 0$, so $I + A$ is invertible.

1.7 proof:

Using $X^H = -X$,

$$U^H = ((I - X)^{-1})^H (I + X)^H = (I + X)^{-1}(I - X),$$

so

$$U^H U = (I + X)^{-1}(I - X)(I + X)(I - X)^{-1} = I.$$

2. (20 points)

For each matrix, compute both the projection form and the diagonalization form.

Matrix A. Let

$$A = \begin{pmatrix} -1 & 6 \\ -3 & 8 \end{pmatrix}.$$

(a) Find the characteristic polynomial $\det(tI - A)$.

(b) Find the spectral decomposition

$$A = \lambda_1 P_1 + \lambda_2 P_2,$$

where P_1 and P_2 are spectral projections. They are not required to be orthogonal projections.

(c) Find a diagonalization $A = PDP^{-1}$. Explicitly write P, D, and P^{-1}.

(d) Use either the spectral decomposition or diagonalization to compute A^{10}. Your final answer must be written as an entry-wise matrix. Leaving the answer as a product of matrices receives no credit.

Matrix B. Let

$$B = \begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

(e) Find the characteristic polynomial $\det(tI - B)$.

(f) Find the spectral decomposition of B

$$B = \lambda_1 P_1 + \lambda_2 P_2,$$

where P_1 and P_2 are orthogonal projections.

(g) Diagonalize B by an orthogonal matrix:

$$B = QDQ^T.$$

Explicitly write Q, D, and Q^T.

Solution: **Problem 2, Matrix A.**

$$\det(tI - A) = \det \begin{pmatrix} t+1 & -6 \\ 3 & t-8 \end{pmatrix} = (t-2)(t-5).$$

The Lagrange spectral projections are

$$P_2 = \frac{A - 5I}{2-5} = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}, \qquad P_5 = \frac{A - 2I}{5-2} = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}.$$

The spectral form, cross-filled projection form, and diagonalization form are one equation:

$$\begin{aligned} A = 2P_2 + 5P_5 &= 2\underbrace{\begin{pmatrix} \mathbf{2} & -2 \\ 1 & -1 \end{pmatrix}}_{P_2} + 5\underbrace{\begin{pmatrix} -1 & 2 \\ -1 & \mathbf{2} \end{pmatrix}}_{P_5} \\ &= 2\left(\begin{pmatrix} \mathbf{2} \\ 1 \end{pmatrix} \frac{1}{\mathbf{2}} \begin{pmatrix} \mathbf{2} & -2 \end{pmatrix}\right) + 5\left(\begin{pmatrix} 2 \\ \mathbf{2} \end{pmatrix} \frac{1}{\mathbf{2}} \begin{pmatrix} -1 & \mathbf{2} \end{pmatrix}\right) \\ &= 2\left(\begin{pmatrix} 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}\right) + 5\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \end{pmatrix}\right) \\ &= \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}}_{P^{-1}}. \end{aligned}$$

Using the spectral decomposition,

$$A^{10} = 2^{10} P_2 + 5^{10} P_5 = \begin{pmatrix} -9763577 & 19529202 \\ -9764601 & 19530226 \end{pmatrix}.$$

Problem 2, Matrix B.

$$\det(tI - B) = (t-5)(t-2)^2.$$

The spectral projections are

$$P_5 = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad P_2 = I - P_5 = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

The spectral form, diagonal cross-filling form, square-root distribution, and orthogonal diagonalization form are one equation:

$$\begin{aligned} B &= 2P_2 + 5P_5 \\ &= 2\underbrace{\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}}_{P_2} + 5\underbrace{\begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}}_{P_5} \end{aligned}$$

$$
\begin{aligned}
&= 2\left(\underbrace{\begin{pmatrix} \mathbf{\frac{2}{3}} & -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} \\ -\frac{1}{3} & \frac{1}{6} & \frac{1}{6} \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{\frac{1}{2}} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{R_2}\right) + 5\underbrace{\begin{pmatrix} \mathbf{\frac{1}{3}} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}}_{R_3} \\
&= 2\left(\begin{pmatrix} \mathbf{\frac{2}{3}} \\ -\frac{1}{3} \\ -\frac{1}{3} \end{pmatrix} \frac{1}{\mathbf{\frac{2}{3}}} \begin{pmatrix} \mathbf{\frac{2}{3}} & -\frac{1}{3} & -\frac{1}{3} \end{pmatrix} + \begin{pmatrix} 0 \\ \mathbf{\frac{1}{2}} \\ -\frac{1}{2} \end{pmatrix} \frac{1}{\mathbf{\frac{1}{2}}} \begin{pmatrix} 0 & \mathbf{\frac{1}{2}} & -\frac{1}{2} \end{pmatrix}\right) + 5\left(\begin{pmatrix} \mathbf{\frac{1}{3}} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} \frac{1}{\mathbf{\frac{1}{3}}} \begin{pmatrix} \mathbf{\frac{1}{3}} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}\right) \\
&= 2\left(\begin{pmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ -\frac{1}{3} \end{pmatrix} \frac{1}{\sqrt{\mathbf{\frac{2}{3}}}} \frac{1}{\sqrt{\mathbf{\frac{2}{3}}}} \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} \frac{1}{\sqrt{\mathbf{\frac{1}{2}}}} \frac{1}{\sqrt{\mathbf{\frac{1}{2}}}} \begin{pmatrix} 0 & \frac{1}{2} & -\frac{1}{2} \end{pmatrix}\right) + 5\left(\begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{pmatrix} \frac{1}{\sqrt{\mathbf{\frac{1}{3}}}} \frac{1}{\sqrt{\mathbf{\frac{1}{3}}}} \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}\right) \\
&= 2\left(\begin{pmatrix} \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}\right) + 5\left(\begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}\right) \\
&= \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}}_{Q} \underbrace{\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}}_{Q^T}.
\end{aligned}
$$

3. (15 points)

Consider the matrix

$$A = \begin{pmatrix} 2 & * & 4 \\ * & 5 & * \\ -1 & -2 & * \end{pmatrix},$$

where the four entries marked $*$ are unknown. Suppose A is diagonalizable and

$$\det(tI - A) = t^2(t - \lambda).$$

(a) Find λ.

(b) Complete the matrix by determining all entries marked $*$.

(c) Find the spectral decomposition of A.

(d) Find a diagonalization $A = PDP^{-1}$. Explicitly write P, D, and P^{-1}.

Solution: **Problem 3.** Since the matrix is diagonalizable and

$$\det(tI - A) = t^2(t - \lambda),$$

there is a spectral decomposition

$$A = 0 \cdot P_0 + \lambda P_\lambda = \lambda P_\lambda.$$

The projection P_λ has rank one, so A is a multiple of one rank-one projection. Hence A has rank one and can be completed by cross-filling from the pivot $a_{11} = 2$:

$$A = \frac{1}{2}\begin{pmatrix} 2 \\ \frac{5}{2} \\ -1 \end{pmatrix}\begin{pmatrix} 2 & 4 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 & 4 \\ \frac{5}{2} & 5 & 5 \\ -1 & -2 & -2 \end{pmatrix}.$$

Therefore

$$\lambda = \operatorname{tr}(A) = 2 + 5 - 2 = 5.$$

The spectral projections are

$$P_5 = \frac{1}{5}A = \begin{pmatrix} \frac{2}{5} & \frac{4}{5} & \frac{4}{5} \\ \frac{1}{2} & 1 & 1 \\ -\frac{1}{5} & -\frac{2}{5} & -\frac{2}{5} \end{pmatrix}, \qquad P_0 = I - P_5 = \begin{pmatrix} \frac{3}{5} & -\frac{4}{5} & -\frac{4}{5} \\ -\frac{1}{2} & 0 & -1 \\ \frac{1}{5} & \frac{2}{5} & \frac{7}{5} \end{pmatrix}.$$

The spectral form, cross-filled projection form, scalar distribution, and diagonalization form are one equation:

$$A = 5P_5 + 0P_0$$

$$= 5 \underbrace{\begin{pmatrix} \frac{2}{5} & \frac{4}{5} & \frac{4}{5} \\ \frac{1}{2} & 1 & 1 \\ -\frac{1}{5} & -\frac{2}{5} & -\frac{2}{5} \end{pmatrix}}_{P_5} + 0 \underbrace{\begin{pmatrix} \frac{3}{5} & -\frac{4}{5} & -\frac{4}{5} \\ -\frac{1}{2} & 0 & -1 \\ \frac{1}{5} & \frac{2}{5} & \frac{7}{5} \end{pmatrix}}_{P_0}$$

$$= 5 \underbrace{\begin{pmatrix} \frac{2}{5} & \frac{4}{5} & \frac{4}{5} \\ \frac{1}{2} & \mathbf{1} & 1 \\ -\frac{1}{5} & -\frac{2}{5} & -\frac{2}{5} \end{pmatrix}}_{P_5} + 0 \left(\underbrace{\begin{pmatrix} \mathbf{\frac{3}{5}} & -\frac{4}{5} & -\frac{4}{5} \\ -\frac{1}{2} & \frac{2}{3} & \frac{2}{3} \\ \frac{1}{5} & -\frac{4}{15} & -\frac{4}{15} \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & -\frac{2}{3} & -\frac{5}{3} \\ 0 & \mathbf{\frac{2}{3}} & \frac{5}{3} \end{pmatrix}}_{R_2} \right)$$

$$= 5 \left(\begin{pmatrix} \frac{4}{5} \\ \mathbf{1} \\ -\frac{2}{5} \end{pmatrix} \frac{1}{\mathbf{1}} \begin{pmatrix} \frac{1}{2} & \mathbf{1} & 1 \end{pmatrix} \right) + 0 \left(\begin{pmatrix} \mathbf{\frac{3}{5}} \\ -\frac{1}{2} \\ \frac{1}{5} \end{pmatrix} \frac{1}{\mathbf{\frac{3}{5}}} \begin{pmatrix} \mathbf{\frac{3}{5}} & -\frac{4}{5} & -\frac{4}{5} \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{2}{3} \\ \mathbf{\frac{2}{3}} \end{pmatrix} \frac{1}{\mathbf{\frac{2}{3}}} \begin{pmatrix} 0 & \mathbf{\frac{2}{3}} & \frac{5}{3} \end{pmatrix} \right)$$

$$= 5 \left(\begin{pmatrix} \frac{4}{5} \\ 1 \\ -\frac{2}{5} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 1 & 1 \end{pmatrix} \right) + 0 \left(\begin{pmatrix} \frac{3}{5} \\ -\frac{1}{2} \\ \frac{1}{5} \end{pmatrix} \begin{pmatrix} 1 & -\frac{4}{3} & -\frac{4}{3} \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 & \frac{5}{2} \end{pmatrix} \right)$$

$$= \underbrace{\begin{pmatrix} \frac{3}{5} & 0 & \frac{4}{5} \\ -\frac{1}{2} & -\frac{2}{3} & 1 \\ \frac{1}{5} & \frac{2}{3} & -\frac{2}{5} \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & -\frac{4}{3} & -\frac{4}{3} \\ 0 & 1 & \frac{5}{2} \\ \frac{1}{2} & 1 & 1 \end{pmatrix}}_{P^{-1}}.$$

4. (20 points)

Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Find the characteristic polynomial $\det(tI - A^T A)$.

(b) Find the singular values of A.

(c) Find a singular value decomposition $A = U\Sigma V^T$.

Solution: **Problem 4.**

$$A^T A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \det(tI - A^T A) = (t-3)(t-1).$$

The singular values are

$$\sigma_1 = \sqrt{3}, \qquad \sigma_2 = 1.$$

By Lagrange interpolation,

$$P_3 = \frac{A^T A - I}{3-1} = \frac{1}{2}\left(\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix},$$

$$P_1 = \frac{A^T A - 3I}{1-3} = -\frac{1}{2}\left(\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \right) = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

The spectral decomposition of $A^T A$, diagonal cross-filling, square-root distribution, and orthogonal diagonalization are one equation:

$$A^T A = 3P_3 + P_1 = 3 \underbrace{\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_3} + \underbrace{\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_1}$$

$$= 3 \underbrace{\begin{pmatrix} \mathbf{\frac{1}{2}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}}_{P_3} + \underbrace{\begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \mathbf{\frac{1}{2}} \end{pmatrix}}_{P_1}$$

$$= 3 \left(\begin{pmatrix} \mathbf{\frac{1}{2}} \\ \frac{1}{2} \end{pmatrix} \frac{1}{\mathbf{\frac{1}{2}}} \begin{pmatrix} \mathbf{\frac{1}{2}} & \frac{1}{2} \end{pmatrix} \right) + \left(\begin{pmatrix} -\frac{1}{2} \\ \mathbf{\frac{1}{2}} \end{pmatrix} \frac{1}{\mathbf{\frac{1}{2}}} \begin{pmatrix} -\frac{1}{2} & \mathbf{\frac{1}{2}} \end{pmatrix} \right)$$

$$= 3\left(\begin{pmatrix}\frac{1}{2}\\ \frac{1}{2}\end{pmatrix}\frac{1}{\sqrt{\mathbf{\frac{1}{2}}}}\frac{1}{\sqrt{\mathbf{\frac{1}{2}}}}\begin{pmatrix}\frac{1}{2} & \frac{1}{2}\end{pmatrix}\right) + \left(\begin{pmatrix}-\frac{1}{2}\\ \frac{1}{2}\end{pmatrix}\frac{1}{\sqrt{\mathbf{\frac{1}{2}}}}\frac{1}{\sqrt{\mathbf{\frac{1}{2}}}}\begin{pmatrix}-\frac{1}{2} & \frac{1}{2}\end{pmatrix}\right)$$

$$= 3\left(\begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\right) + \left(\begin{pmatrix}-\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix}\begin{pmatrix}-\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}\right)$$

$$= \underbrace{\begin{pmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}}_{V}\underbrace{\begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix}}_{\Lambda}\underbrace{\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}}_{V^T}.$$

$$A\underbrace{\begin{pmatrix}\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}}_{V} = \underbrace{\begin{pmatrix}\frac{2}{\sqrt{6}} & 0\\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}}\end{pmatrix}}_{U_2}\underbrace{\begin{pmatrix}\sqrt{3} & 0\\ 0 & 1\end{pmatrix}}_{\Sigma_{2\times 2}}.$$

The remaining output column is also obtained by diagonal cross-filling:

$$I - U_2U_2^T = \underbrace{\begin{pmatrix}\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3}\\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\end{pmatrix}}_{P_0}$$

$$= \underbrace{\begin{pmatrix}\mathbf{\frac{1}{3}} & -\frac{1}{3} & -\frac{1}{3}\\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\\ -\frac{1}{3} & \frac{1}{3} & \frac{1}{3}\end{pmatrix}}_{P_0} = \begin{pmatrix}\mathbf{\frac{1}{3}}\\ -\frac{1}{3}\\ -\frac{1}{3}\end{pmatrix}\frac{1}{\mathbf{\frac{1}{3}}}\begin{pmatrix}\mathbf{\frac{1}{3}} & -\frac{1}{3} & -\frac{1}{3}\end{pmatrix}$$

$$= \begin{pmatrix}\frac{1}{3}\\ -\frac{1}{3}\\ -\frac{1}{3}\end{pmatrix}\frac{1}{\sqrt{\mathbf{\frac{1}{3}}}}\frac{1}{\sqrt{\mathbf{\frac{1}{3}}}}\begin{pmatrix}\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3}\end{pmatrix} = \begin{pmatrix}\frac{1}{\sqrt{3}}\\ -\frac{1}{\sqrt{3}}\\ -\frac{1}{\sqrt{3}}\end{pmatrix}\begin{pmatrix}\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}}\end{pmatrix}.$$

Thus the SVD is the single factorization

$$A = \underbrace{\begin{pmatrix}\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}}\\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}}\end{pmatrix}}_{U}\underbrace{\begin{pmatrix}\sqrt{3} & 0\\ 0 & 1\\ 0 & 0\end{pmatrix}}_{\Sigma}\underbrace{\begin{pmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{pmatrix}}_{V^T}.$$

5. (20 points)

Solve the system

$$\mathbf{x}' = A\mathbf{x}, \qquad \mathbf{x}(0) = \begin{pmatrix}1\\0\\2\end{pmatrix}, \qquad A = \begin{pmatrix}7 & -6 & 10\\ 19 & -13 & 20\\ 6 & -6 & 11\end{pmatrix}.$$

This matrix has one integer eigenvalue and one complex-conjugate pair of eigenvalues with integer real and imaginary parts. It is not triangular or block triangular, so the problem cannot be solved by simply reading off a visible block.

(a) Find the characteristic polynomial $\det(tI - A)$.

(b) Find all eigenvalues of A.

(c) Find the spectral decomposition of A over $\mathbb{C}$.

(d) Use the spectral decomposition to write a formula for e^{At} in real form using exponentials, sine, and cosine. Then solve the initial value problem by computing

$$\mathbf{x}(t) = e^{At}\begin{pmatrix}1\\0\\2\end{pmatrix}.$$

The final answers for both e^{At} and $\mathbf{x}(t)$ must be written entry-wise. Leaving e^{At} only as a product or sum of matrices receives no credit.

(e) Describe the behavior of $\mathbf{x}(t)$ as $t \to \infty$. Your answer should distinguish the real eigenvalue mode from the oscillatory complex-eigenvalue modes.

Solution: Problem 5.

$$\det(tI - A) = (t-1)(t^2 - 4t + 13),$$

so the eigenvalues are

$$1, \qquad 2+3i, \qquad 2-3i.$$

First find the projection for the real eigenvalue:

$$P_1 = \frac{(A-(2+3i)I)(A-(2-3i)I)}{(1-(2+3i))(1-(2-3i))} = \frac{(A-2I)^2+9I}{10} = \begin{pmatrix}-2 & 0 & 2\\ -7 & 0 & 7\\ -3 & 0 & 3\end{pmatrix},$$

so

$$P_{2+3i} + P_{2-3i} = I - P_1.$$

Write

$$P_{2+3i} = X + iY, \qquad P_{2-3i} = X - iY.$$

Their sum is the remaining part of the identity:

$$P_{2+3i} + P_{2-3i} = (X+iY) + (X-iY) = 2X = I - P_1.$$

When A acts on these two projections,

$$AP_{2+3i} = (2+3i)P_{2+3i}, \qquad AP_{2-3i} = (2-3i)P_{2-3i}.$$

Thus the complex part of A is

$$\begin{aligned} A - P_1 &= (2+3i)P_{2+3i} + (2-3i)P_{2-3i} \\ &= (2+3i)(X+iY) + (2-3i)(X-iY) = 4X - 6Y. \end{aligned}$$

Therefore

$$X = \frac{1}{2}(I - P_1), \qquad Y = \frac{1}{6}(P_1 + 4X - A).$$

This gives

$$X = \begin{pmatrix}\frac{3}{2} & 0 & -1\\ \frac{7}{2} & \frac{1}{2} & -\frac{7}{2}\\ \frac{3}{2} & 0 & -1\end{pmatrix}, \qquad Y = \begin{pmatrix}-\frac{1}{2} & 1 & -2\\ -2 & \frac{5}{2} & -\frac{9}{2}\\ -\frac{1}{2} & 1 & -2\end{pmatrix}.$$

Equivalently, the complex spectral projections are

$$P_{2+3i} = X + iY, \qquad P_{2-3i} = X - iY.$$

Therefore

$$e^{At} = e^t P_1 + e^{2t}\big(\cos(3t)(2X) + \sin(3t)(-2Y)\big).$$

Entry-wise,

$$e^{At} = \begin{pmatrix} e^{2t}(3\cos 3t - \sin 3t) - 2e^t & 2e^{2t}\sin 3t & -2e^{2t}(\cos 3t + 2\sin 3t) + 2e^t\\ e^{2t}(7\cos 3t - 4\sin 3t) - 7e^t & e^{2t}(\cos 3t + 5\sin 3t) & -e^{2t}(7\cos 3t + 9\sin 3t) + 7e^t\\ e^{2t}(3\cos 3t - \sin 3t) - 3e^t & 2e^{2t}\sin 3t & -2e^{2t}(\cos 3t + 2\sin 3t) + 3e^t \end{pmatrix}.$$

Thus

$$\mathbf{x}(t) = \begin{pmatrix} e^{2t}(-\cos 3t - 9\sin 3t) + 2e^t\\ e^{2t}(-7\cos 3t - 22\sin 3t) + 7e^t\\ e^{2t}(-\cos 3t - 9\sin 3t) + 3e^t \end{pmatrix}.$$

As $t \to \infty$, the complex pair contributes oscillatory terms of size e^{2t}, while the real eigenvalue contributes a non-oscillatory term of size e^{t}. Hence the oscillatory complex modes dominate.

6. (10 points)

(a) Determine whether

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

is positive definite, positive semidefinite but not positive definite, or neither. If it is positive semidefinite, write

$$B = MM^T$$

for some real matrix M.

(b) Prove: if A is an $n \times n$ positive definite matrix and C is an $m \times n$ matrix with $\operatorname{rank}(C) = n$, then $C^T AC$ is positive definite.

(c) Let

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$$

from Problem 2. Find a positive definite matrix B such that $B^2 = A$.

Solution: Problem 6. For

$$B = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix},$$

diagonal cross-filling gives the following long factorization:

$$B = \underbrace{\begin{pmatrix} \mathbf{2} & 1 & 0 \\ 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{\frac{5}{2}} & 1 \\ 0 & 1 & \frac{2}{5} \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{\frac{8}{5}} \end{pmatrix}}_{R_3}$$

$$= \begin{pmatrix} \mathbf{2} \\ 1 \\ 0 \end{pmatrix} \frac{1}{\mathbf{2}} \begin{pmatrix} \mathbf{2} & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \mathbf{\frac{5}{2}} \\ 1 \end{pmatrix} \frac{1}{\mathbf{\frac{5}{2}}} \begin{pmatrix} 0 & \mathbf{\frac{5}{2}} & 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \mathbf{\frac{8}{5}} \end{pmatrix} \frac{1}{\mathbf{\frac{8}{5}}} \begin{pmatrix} 0 & 0 & \mathbf{\frac{8}{5}} \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \frac{1}{\sqrt{\mathbf{2}}} \frac{1}{\sqrt{\mathbf{2}}} \begin{pmatrix} 2 & 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \frac{5}{2} \\ 1 \end{pmatrix} \frac{1}{\sqrt{\mathbf{\frac{5}{2}}}} \frac{1}{\sqrt{\mathbf{\frac{5}{2}}}} \begin{pmatrix} 0 & \frac{5}{2} & 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{8}{5} \end{pmatrix} \frac{1}{\sqrt{\mathbf{\frac{8}{5}}}} \frac{1}{\sqrt{\mathbf{\frac{8}{5}}}} \begin{pmatrix} 0 & 0 & \frac{8}{5} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{2} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{\frac{5}{2}} \\ \sqrt{\frac{2}{5}} \end{pmatrix} \begin{pmatrix} 0 & \sqrt{\frac{5}{2}} & \sqrt{\frac{2}{5}} \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \sqrt{\frac{8}{5}} \end{pmatrix} \begin{pmatrix} 0 & 0 & \sqrt{\frac{8}{5}} \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} \sqrt{2} & 0 & 0 \\ \frac{1}{\sqrt{2}} & \sqrt{\frac{5}{2}} & 0 \\ 0 & \sqrt{\frac{2}{5}} & \sqrt{\frac{8}{5}} \end{pmatrix}}_{M} \underbrace{\begin{pmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \sqrt{\frac{5}{2}} & \sqrt{\frac{2}{5}} \\ 0 & 0 & \sqrt{\frac{8}{5}} \end{pmatrix}}_{M^T}.$$

Thus the pivots are

$$2, \qquad \frac{5}{2}, \qquad \frac{8}{5},$$

so B is positive definite. For part (b), for $x \neq 0$,

$$x^T C^T ACx = (Cx)^T A(Cx).$$

Since $\operatorname{rank}(C) = n$, the columns of C are independent, so $Cx \neq 0$ for $x \neq 0$. Because A is positive definite, $(Cx)^T A(Cx) > 0$. Hence $C^T AC$ is positive definite.

For

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix},$$

we have eigenvalues $6, 1$ and spectral projections

$$P_6 = \begin{pmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix}, \qquad P_1 = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{pmatrix}.$$

Thus the positive square root is written entrywise as

$$B = \sqrt{6}\,P_6 + P_1 = \sqrt{6}\underbrace{\begin{pmatrix} \frac{4}{5} & \frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix}}_{P_6} + \underbrace{\begin{pmatrix} \frac{1}{5} & -\frac{2}{5} \\ -\frac{2}{5} & \frac{4}{5} \end{pmatrix}}_{P_1} = \begin{pmatrix} \frac{1+4\sqrt{6}}{5} & \frac{-2+2\sqrt{6}}{5} \\ \frac{-2+2\sqrt{6}}{5} & \frac{4+\sqrt{6}}{5} \end{pmatrix}.$$

Then B is positive definite and $B^2 = A$.

7. (10 points)

Let A be a real $n \times n$ matrix, diagonalizable over $\mathbb{C}$. In the spectral decomposition

$$A = \sum \lambda_j P_j,$$

suppose $\lambda = a + bi$ with $b \neq 0$ is a complex eigenvalue with eigenprojection $P = X + Yi$, where X, Y are real $n \times n$ matrices. Since A is real, $\bar{\lambda} = a - bi$ is also an eigenvalue with eigenprojection $\bar{P} = X - Yi$.

You may use the following properties without proof:

$$P^2 = P, \qquad P\bar{P} = 0.$$

Prove the following:

(a) $XY = YX$.

(b) $2X$ is a projection; that is, $(2X)^2 = 2X$.

(c) $Y^2 = -\dfrac{1}{2}X$.

(d) $2XY = Y$.

(e) $\mathrm{Col}(Y) \subseteq \mathrm{Col}(X)$.

Solution: Problem 7. Write $P = X + iY$ and $\bar{P} = X - iY$. From $P^2 = P$,

$$(X + iY)^2 = X + iY.$$

Comparing real and imaginary parts gives

$$X^2 - Y^2 = X, \qquad XY + YX = Y.$$

From $P\bar{P} = 0$,

$$(X + iY)(X - iY) = 0.$$

Comparing real and imaginary parts gives

$$X^2 + Y^2 = 0, \qquad YX - XY = 0.$$

Therefore $XY = YX$. Combining

$$X^2 - Y^2 = X, \qquad X^2 + Y^2 = 0$$

gives

$$2X^2 = X,$$

so

$$(2X)^2 = 4X^2 = 2X.$$

Thus $2X$ is a projection. Also

$$Y^2 = -X^2 = -\frac{1}{2}X.$$

Since $XY = YX$, the equation $XY + YX = Y$ becomes

$$2XY = Y.$$

Finally, if $y \in \mathrm{Col}(Y)$, then $y = Yv$ for some v. Since $Y = 2XY$,

$$y = Yv = 2XYv = X(2Yv) \in \mathrm{Col}(X).$$

Hence $\mathrm{Col}(Y) \subseteq \mathrm{Col}(X)$.